

Self-Check Questions: OpenStax Calculus

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1 Calculus Vol2 - 2. Applications of Integration

<https://openstax.org/books/calculus-volume-2/pages/2-introduction>

1.1 Calculus Vol2 - 2.4 Arc Length of a Curve and Surface Area

<https://openstax.org/books/calculus-volume-2/pages/2-4-arc-length-of-a-curve-and-surface-area>

1. How can the length of a curve be approximated using line segments?
2. How does partitioning an interval lead to a Riemann sum for arc length?
3. What is the formula for the arc length of a curve $y = f(x)$ on $[a, b]$?
4. What assumptions must be satisfied by $f(x)$ for the arc length formula to apply?
5. Why does the arc length formula involve $\sqrt{1 + [f'(x)]^2}$?
6. How is the Pythagorean theorem used in deriving the arc length formula?
7. How do you compute arc length when the curve is given as $x = g(y)$?
8. How does the derivation of arc length for $x = g(y)$ differ from that for $y = f(x)$?
9. What role do smoothness and differentiability play in arc length problems?
10. How can you decide whether to integrate with respect to x or y ?
11. What are common substitutions used when evaluating arc length integrals?
12. How can symmetry simplify arc length computations?
13. What is a surface of revolution?
14. How can a surface of revolution be approximated using frustums of cones?
15. What is a frustum, and why is it useful in this context?
16. How does approximating a surface with frustums lead to a Riemann sum?
17. How is the surface area formula derived from arc length ideas?
18. What is the formula for the surface area when rotating $y = f(x)$ about the x -axis?
19. What is the formula for the surface area when rotating $x = f(y)$ about the y -axis?
20. Why does the surface area formula include a factor of 2π ?
21. What geometric quantity does the factor $2\pi f(x)$ or $2\pi x$ represent?
22. How does arc length appear inside the surface area integral?
23. How do you modify the surface area formula when the curve is given as $x = g(y)$?
24. What similarities exist between arc length and surface area formulas?

1.2 Calculus Vol2 - 2.5 Physical Applications

<https://openstax.org/books/calculus-volume-2/pages/2-5-physical-applications>

1. What is a density function, and how can it vary along a rod or region?
2. How do you compute the mass of a thin rod with variable linear density?
3. What is the formula for mass in terms of an integral over an interval?
4. How does the concept of a Riemann sum apply to mass calculations?
5. How is area density different from linear density?
6. How do you compute the mass of a two-dimensional region with variable density?
7. What is radial density, and in what situations is it used?
8. How does symmetry simplify mass calculations in physical problems?
9. What is work in the context of physics?
10. How is work defined in terms of force and displacement?
11. How can work be approximated using sums of small contributions?
12. What is the general formula for work done by a variable force?
13. How do you compute work when the force depends on position?
14. What is Hooke's Law, and how does it relate force to displacement?
15. How is work computed when stretching or compressing a spring?
16. Why does the integral for spring work involve the force function $F(x) = kx$?
17. How do you set up limits of integration in spring problems?
18. What is a pumping problem in calculus?
19. How is work computed when lifting a fluid (e.g., water) from a tank?
20. How do you model a fluid as a collection of thin horizontal slices?
21. What quantities must be determined for each slice in a pumping problem?
22. How do you determine the distance each slice must be lifted?
23. How does density of a fluid affect the work calculation?
24. What is the role of weight density in fluid problems?
25. How do you express the work required to pump fluid as an integral?
26. What is hydrostatic pressure?
27. How is pressure related to depth in a fluid?
28. What units are used for pressure?
29. How do you compute the force exerted by a fluid on a submerged surface?
30. How is force related to pressure and area?
31. How do you model a submerged surface using horizontal strips?
32. How do you compute the force on each strip?
33. How is the total force found using an integral?

34. What assumptions are made about fluid density in hydrostatic problems?
35. How can changing the shape of a container affect work or force calculations?

1.3 Calculus Vol2 - 2.6 Moments and Centers of Mass

<https://openstax.org/books/calculus-volume-2/pages/2-6-moments-and-centers-of-mass>

1. What is the definition of the center of mass of a system?
2. How can the center of mass be interpreted physically as a “balancing point”?
3. What is a moment of a system of masses with respect to a point (e.g., the origin)?
4. For point masses on a line, how do you compute the total mass?
5. For point masses on a line, how do you compute the moment with respect to the origin?
6. How do you compute the center of mass $\bar{x}$ for discrete point masses on a line?
7. How does the formula for center of mass relate to a weighted average?
8. For point masses in the plane, how are the moments M_x and M_y defined?
9. How do you compute the coordinates $(\bar{x}, \bar{y})$ of the center of mass in the plane?
10. Why does the center of mass depend more heavily on regions with larger mass?
11. What changes when moving from discrete masses to a continuous mass distribution?
12. What is a lamina?
13. What assumption is often made about density when studying laminas?
14. How is mass computed for a continuous distribution along a line?
15. How is mass computed for a lamina bounded by a function $f(x)$ on $[a, b]$?
16. How do you define the moment of a continuous mass distribution using integrals?
17. How are M_x and M_y computed for a lamina using integrals?
18. How do you compute the center of mass $(\bar{x}, \bar{y})$ for a lamina?
19. What is the centroid of a region?
20. When does the centroid coincide with the center of mass?
21. Why does constant density simplify center of mass calculations?
22. How can symmetry be used to determine the center of mass without integration?
23. Where is the center of mass of a rectangular lamina located?
24. What role does partitioning (Riemann sums) play in deriving formulas for mass and moments?
25. How do integrals arise naturally from approximating a lamina by small rectangles?
26. What is the geometric meaning of M_x and M_y ?
27. Why is the center of mass independent of density when density is constant?
28. How would variable density change the formulas for mass and moments?
29. How does the concept of center of mass connect to real-world applications (e.g., balance, engineering)?
30. What is the relationship between center of mass and equilibrium?
31. How can you check whether your computed center of mass is reasonable using symmetry or intuition?

32. How does the idea of “balancing point” generalize from 1D to 2D regions?
33. What are the key steps to solve a typical center of mass problem for a region?
34. What are the key differences between discrete and continuous mass systems?
35. How can you set up integrals for regions bounded by curves when finding centers of mass?

1.4 Calculus Vol2 - 2.7 Integrals, Exponential Functions, and Logarithms

<https://openstax.org/books/calculus-volume-2/pages/2-7-integrals-exponential-functions-and-logarithms>

1. How can the natural logarithm function $\ln x$ be defined using a definite integral?
2. Why does the usual power rule for integration fail for $\int \frac{1}{x}, dx$, and how does the integral definition of $\ln x$ resolve this?
3. What is the domain of the function $\ln x$ when defined via an integral, and why?
4. How is $\ln x = \int_1^x \frac{1}{t}, dt$ interpreted geometrically?
5. What is the value of $\ln 1$, and how does it follow from the integral definition?
6. Is $\ln x$ increasing or decreasing on its domain? How can this be seen from the integral definition?
7. What is the derivative of $\ln x$, and how does it follow from the Fundamental Theorem of Calculus?
8. How can you use the definition of $\ln x$ to show it is continuous and differentiable?
9. How is the number e defined using the natural logarithm?
10. What special property defines the number e in terms of $\ln x$?
11. How is the exponential function e^x defined as an inverse function?
12. What is the derivative of e^x , and why is this property significant?
13. What is the integral of e^x ?
14. How do you compute $\int e^{kx}, dx$ for a constant k ?
15. How are general exponential functions a^x expressed in terms of e^x ?
16. What is the formula for $\int a^x, dx$ and how does $\ln a$ appear in it?
17. How are general logarithmic functions $\log_a x$ expressed using $\ln x$?
18. What is the derivative of $\log_a x$ in terms of $\ln a$?
19. What are the key logarithmic properties (product, quotient, power rules), and how can they be justified using integrals?
20. How can you prove that $\ln(ab) = \ln a + \ln b$ using the integral definition?
21. How can you prove that $\ln(a^r) = r \ln a$?

2 Calculus Vol2 - 4. Introduction to Differential Equations

<https://openstax.org/books/calculus-volume-2/pages/4-introduction>

2.1 Calculus Vol2 - 4.1 Basics of Differential Equations

<https://openstax.org/books/calculus-volume-2/pages/4-1-basics-of-differential-equations>

1. What is a differential equation?
2. What is meant by a solution to a differential equation?
3. How can you verify that a function $y = f(x)$ is a solution to a given differential equation?
4. Give an example of a differential equation and one of its solutions.
5. Why are solutions to differential equations often not unique?
6. What property of derivatives explains the non-uniqueness of solutions?
7. What is the order of a differential equation?
8. How do you determine the order of a differential equation from its expression?
9. What is the order of the differential equation $y' - 4y = x^2 - 3x + 4$?
10. What is the order of the differential equation $x^2y''' - 3xy'' + xy' - 3y = \sin x$?
11. What is a general solution of a differential equation?
12. Why does a general solution typically include an arbitrary constant C ?
13. For the differential equation $y' = 2x$, what is its general solution?
14. What does the family of functions $y = x^2 + C$ represent?
15. What is a particular solution of a differential equation?
16. How do you obtain a particular solution from a general solution?
17. What is an initial-value problem?
18. How does an initial condition help determine a unique solution?
19. Solve the initial-value problem:

$$y' = 2x, \quad y(2) = 7.$$

20. What is the difference between a general solution and a particular solution?
21. What does it mean for a function to satisfy a differential equation?
22. Verify that $y = e^{-3x} + 2x + 3$ is a solution of

$$y' + 3y = 6x + 11.$$

23. Verify that $y = 2e^{3x} - 2x - 2$ is a solution of

$$y' - 3y = 6x + 4.$$

24. When checking a solution, why must both y and its derivatives be substituted into the equation?

25. Can two different functions be solutions to the same differential equation? Explain why.
26. What role does the constant C play geometrically in the family of solutions?
27. How would the graph of $y = x^2 + C$ change as C varies?
28. Why is identifying the order of a differential equation useful?
29. What information is needed in addition to a differential equation to determine a unique solution?

2.2 Calculus Vol2 - 4.2 Direction Fields and Numerical Methods

<https://openstax.org/books/calculus-volume-2/pages/4-2-direction-fields-and-numerical-methods>

1. What is a direction field (slope field) for a differential equation?
2. For what type of differential equations are direction fields typically used?
3. What is the general form of a first-order differential equation used in direction fields?
4. What does each small line segment in a direction field represent?
5. How is the slope of a line segment at a point (x, y) determined?
6. How do you construct a direction field for a differential equation $y' = f(x, y)$?
7. Why can a direction field be drawn without solving the differential equation explicitly?
8. What qualitative information can a direction field provide about solutions?
9. What is a solution curve in the context of a direction field?
10. How can you sketch a solution curve using a direction field?
11. Why do solution curves follow the line segments in a direction field?
12. What is an initial condition in the context of direction fields?
13. How do you use an initial point (x_0, y_0) to sketch a particular solution?
14. Why do solution curves corresponding to different initial conditions typically not intersect?
15. What is an equilibrium solution?
16. How can equilibrium solutions be identified from a differential equation?
17. How do equilibrium solutions appear in a direction field?
18. What does it mean for an equilibrium solution to be stable?
19. What does it mean for an equilibrium solution to be unstable?
20. How can you determine stability visually from a direction field?
21. What is the purpose of numerical methods in differential equations?
22. Why are numerical methods needed when solving differential equations?
23. What is Euler's Method?
24. What type of problems is Euler's Method used to approximate?
25. What is the idea behind Euler's Method in terms of tangent line approximation?
26. What is the step size h in Euler's Method?
27. How does the choice of step size affect the accuracy of Euler's Method?
28. What is the iterative formula for Euler's Method?
29. Starting from (x_n, y_n) , how do you compute (x_{n+1}, y_{n+1}) ?
30. Write the update rule:
$$y_{n+1} = y_n + hf(x_n, y_n).$$
31. How is x_{n+1} computed from x_n ?

32. Apply Euler's Method to approximate the solution of

$$y' = f(x, y), \quad y(x_0) = y_0$$

for one step.

33. Apply Euler's Method for two steps and express the result explicitly.

34. What geometric idea explains why Euler's Method works?

35. Why does Euler's Method accumulate error over multiple steps?

36. What is the difference between the exact solution and a numerical approximation?

37. How can decreasing the step size h improve the approximation?

38. What is the trade-off when choosing a very small step size?

39. How is Euler's Method related to linear approximation (tangent lines)?

40. In what sense does Euler's Method "follow" the direction field?

41. How would you approximate a solution curve using only a direction field (without formulas)?

42. What are the limitations of direction fields?

43. What are the limitations of Euler's Method?

44. When would you prefer a graphical method over an analytical solution?

45. When would you prefer a numerical method over solving analytically?

46. How can direction fields and Euler's Method be used together?

2.3 Calculus Vol2 - 4.3 Separable Equations

<https://openstax.org/books/calculus-volume-2/pages/4-3-separable-equations>

1. What is a separable differential equation?
2. In what form must a differential equation be written to be considered separable?
3. Write the general form:

$$\frac{dy}{dx} = f(x)g(y).$$

4. Why is such an equation called “separable”?
5. What does it mean to separate variables in a differential equation?
6. How do you rewrite a separable differential equation in differential form?
7. Write the separated form:

$$\frac{dy}{g(y)} = f(x) dx.$$

8. What is the main idea behind the method of separation of variables?
9. What are the steps of the separation of variables method?
10. Why must you check for values where $g(y) = 0$ before separating variables?
11. What do solutions of $g(y) = 0$ represent?
12. Why can constant solutions be lost during the separation process?
13. After separating variables, what is the next step?
14. Why do we integrate both sides of the equation?
15. What form does the equation take after integration?
16. Why is an arbitrary constant C introduced after integration?
17. When solving, why might it not be possible to express y explicitly as a function of x ?
18. What is an implicit solution?
19. Give an example of an implicit solution arising from separation of variables.
20. How do you apply an initial condition to a separable differential equation?
21. What is an initial-value problem in this context?
22. After integrating, how do you solve for the constant using an initial condition?
23. Solve the general structure:

$$\int \frac{1}{g(y)} dy = \int f(x) dx.$$

24. For the equation $y' = (x^2 - 4)(3y + 2)$, how do you separate variables?
25. What constant solution arises from $3y + 2 = 0$?
26. After separation, what integral must be computed:

$$\int \frac{dy}{3y + 2}?$$

27. What substitution is useful for integrating expressions like $\frac{1}{3y+2}$?

28. Why is substitution often needed when integrating the y -side?
29. How do you interpret the final solution after integrating both sides?
30. For the equation $y' = (2x + 3)(y^2 - 4)$, how do you separate variables?
31. What constant solutions arise from $y^2 - 4 = 0$?
32. Why do these constant solutions need to be included separately?
33. What integration technique is typically used for expressions like $\frac{1}{y^2-4}$?
34. How does partial fraction decomposition appear in separable equations?
35. Why are integration techniques from earlier calculus essential here?
36. What is an autonomous differential equation?
37. How can you recognize an autonomous equation from its form?
38. Why is an equation of the form $y' = g(y)$ separable?
39. What is the general solution to an equation of the form $y' = f(x)$?
40. What is the general solution to an equation of the form $y' = g(y)$?
41. How does separation of variables generalize these simpler cases?
42. In what situations is separation of variables especially useful?
43. Why are separable equations common in physics and engineering?
44. What are the limitations of the separation of variables method?
45. Can every differential equation be solved using separation of variables?
46. How can you recognize when a differential equation is not separable?
47. What is the overall strategy when solving separable differential equations?

2.4 Calculus Vol2 - 4.4 The Logistic Equation

<https://openstax.org/books/calculus-volume-2/pages/4-4-the-logistic-equation>

1. What real-world limitation motivates the logistic model of population growth?
2. Why is exponential growth not realistic for large populations?
3. What is the carrying capacity of a population?
4. What does the symbol K represent in the logistic equation?
5. What does the symbol r represent in the logistic equation?
6. Write the logistic differential equation:

$$\frac{dP}{dt} = rP \left(1 - \frac{P}{K} \right).$$

7. Why is the logistic equation nonlinear?
8. What happens to $\frac{dP}{dt}$ when P is very small?
9. How does the logistic equation behave when $P \ll K$?
10. How does the logistic model approximate exponential growth for small populations?
11. What happens to $\frac{dP}{dt}$ when $P = K$?
12. What does it mean physically when $P = K$?
13. What happens to $\frac{dP}{dt}$ when $P > K$?
14. Why does the population decrease when $P > K$?
15. What are the equilibrium solutions of the logistic equation?
16. Solve:

$$rP \left(1 - \frac{P}{K} \right) = 0.$$

17. Why are $P = 0$ and $P = K$ equilibrium solutions?
18. What is meant by stability of an equilibrium solution?
19. Is $P = 0$ stable, unstable, or semi-stable? Explain.
20. Is $P = K$ stable, unstable, or semi-stable? Explain.
21. What is a phase line?
22. How do you construct a phase line for the logistic equation?
23. What does the phase line indicate about the sign of $\frac{dP}{dt}$?
24. For which values of P is the population increasing?
25. For which values of P is the population decreasing?
26. How does the phase line describe long-term behavior of solutions?
27. What happens to $P(t)$ as $t \rightarrow \infty$ when $0 < P_0 < K$?
28. What happens to $P(t)$ as $t \rightarrow \infty$ when $P_0 > K$?
29. Why does the solution approach K but never reach it exactly?

30. What is the general strategy to solve the logistic differential equation?
31. Why is the logistic equation separable?
32. Rewrite the logistic equation in separated form.
33. What algebraic manipulation is needed before integrating?
34. Why is partial fraction decomposition used in solving the logistic equation?
35. What is the general solution of the logistic differential equation?
36. Write the solution:

$$P(t) = \frac{P_0 K e^{rt}}{(K - P_0) + P_0 e^{rt}}.$$

37. What is the role of the initial condition $P(0) = P_0$?
38. How does the solution behave as $t \rightarrow \infty$?
39. How does the solution behave as $t \rightarrow -\infty$?
40. What is the shape of the graph of a logistic function?
41. Why is the logistic curve called sigmoidal (S-shaped)?
42. What is a point of inflection?
43. Why does the logistic solution have a point of inflection?
44. At what population value does the inflection point occur?
45. Why is the growth rate maximal at $P = \frac{K}{2}$?
46. How can you find the inflection point using the second derivative?
47. What happens to concavity before and after the inflection point?
48. How does the logistic model improve on exponential growth models?
49. In what types of real-world systems is the logistic equation used?
50. What assumptions underlie the logistic model?
51. What are the limitations of the logistic equation?

2.5 Calculus Vol2 - 4.5 First-order Linear Equations

<https://openstax.org/books/calculus-volume-2/pages/4-5-first-order-linear-equations>

1. What is a first-order linear differential equation?
2. What is the general (standard) form of a first-order linear differential equation?
3. Write the standard form:

$$y' + p(x)y = q(x).$$

4. How can any first-order linear differential equation be rewritten into standard form?
5. What roles do the functions $p(x)$ and $q(x)$ play?
6. Why is the equation called “linear”?
7. What distinguishes a linear differential equation from a separable one?
8. Can every linear differential equation be solved by separation of variables? Why or why not?
9. What is an integrating factor?
10. Why is an integrating factor useful when solving linear differential equations?
11. What property do we want the integrating factor to create?
12. What is the formula for the integrating factor $\mu(x)$?
13. Write:

$$\mu(x) = e^{\int p(x) dx}.$$

14. How is the integrating factor derived?
 15. What differential equation does $\mu(x)$ satisfy?
 16. Write:
- $$\mu'(x) = \mu(x)p(x).$$
17. After multiplying by $\mu(x)$, what form does the equation take?
 18. Why does the left-hand side become a derivative of a product?
 19. Write the identity:

$$\frac{d}{dx}(\mu(x)y) = \mu(x)y' + \mu(x)p(x)y.$$

20. After multiplying by $\mu(x)$, what equation do we obtain?
21. Write:

$$\frac{d}{dx}(\mu(x)y) = \mu(x)q(x).$$

22. What is the next step after obtaining this form?
23. Why can both sides now be integrated easily?
24. After integrating, what form does the solution take?
25. Write the general solution:

$$\mu(x)y = \int \mu(x)q(x) dx + C.$$

26. How do you solve for $y(x)$ after integration?
27. What is the full step-by-step strategy for solving first-order linear equations?
28. List the five steps of the method.
29. Why is it important to first rewrite the equation in standard form?
30. What happens if the coefficient of y' is not 1?
31. How do you handle an equation of the form $a(x)y' + b(x)y = c(x)$?
32. Why is dividing by $a(x)$ necessary?
33. What is an initial-value problem in this context?
34. How do you apply an initial condition to determine the constant C ?
35. Solve symbolically:

$$y' + p(x)y = 0.$$
36. What is the general solution to the homogeneous equation?
37. How does the homogeneous solution relate to the general solution?
38. What is the difference between homogeneous and nonhomogeneous equations?
39. What happens when $q(x) = 0$?
40. What happens when $q(x) \neq 0$?
41. Why does the solution involve an integral of $\mu(x)q(x)$?
42. What integration techniques are often required in solving these equations?
43. How does the integrating factor method simplify the problem conceptually?
44. What is the geometric interpretation of multiplying by $\mu(x)$?
45. Why does the method always work for first-order linear equations?
46. What are common mistakes when applying the integrating factor method?
47. How can you verify that a solution is correct?
48. In what applications do first-order linear equations appear?
49. How are they used in modeling motion with air resistance?
50. How are they used in electrical circuits?
51. Why are first-order linear equations important in applied mathematics?

3 Calculus Vol2 - 5. Sequences and Series

<https://openstax.org/books/calculus-volume-2/pages/5-introduction>

3.1 Calculus Vol2 - 5.1 Sequences

<https://openstax.org/books/calculus-volume-2/pages/5-1-sequences>

1. What is an infinite sequence? How is it typically denoted?
2. How can a sequence be interpreted as a function? What is its domain?
3. What is the index of a sequence, and what role does it play?
4. What is an explicit formula for a sequence? Give an example.
5. What is a recursive definition of a sequence? How does it differ from an explicit formula?
6. Given a recursive sequence, how can you compute its first few terms?
7. What does it mean to graph a sequence? How does it differ from graphing a function?
8. What is the limit of a sequence?
9. What does it mean for a sequence a_n to converge to a number L ?
10. Write the formal ϵ - N definition of convergence of a sequence.
11. What does it mean for a sequence to diverge?
12. What are some different ways a sequence can diverge?
13. What does it mean for a sequence to diverge to infinity? To negative infinity?
14. Why does changing a finite number of terms of a sequence not affect its convergence?
15. How can limits of functions be used to determine limits of sequences?
16. State the theorem relating $\lim_{x \rightarrow \infty} f(x)$ and $\lim_{n \rightarrow \infty} a_n$ when $a_n = f(n)$.
17. Evaluate $\lim_{n \rightarrow \infty} \frac{1}{n}$ using a function-based argument.
18. What happens to r^n as $n \rightarrow \infty$ when:
 - $0 < r < 1$?
 - $r = 1$?
 - $r > 1$?
19. How can limit laws be applied to sequences?
20. How do you find the limit of a sequence that is a sum or product of simpler sequences?
21. What is a bounded sequence?
22. What does it mean for a sequence to be increasing? Decreasing?
23. What is a monotonic (monotone) sequence?
24. State the Monotone Convergence Theorem.
25. Why is the Monotone Convergence Theorem useful?
26. Give an example of a sequence that is bounded and increasing.
27. Give an example of a sequence that is unbounded.

28. Give an example of a sequence that oscillates and does not converge.
29. How can you determine convergence from the behavior of the terms as $n \rightarrow \infty$?

3.2 Calculus Vol2 - 5.2 Infinite Series

<https://openstax.org/books/calculus-volume-2/pages/5-2-infinite-series>

1. What is an infinite series? How is it related to a sequence?
2. How is an infinite series written using summation notation?
3. What is the difference between a sequence a_n and the series $\sum_{n=1}^{\infty} a_n$?
4. What is a partial sum of a series? How is the k -th partial sum S_k defined?
5. How can an infinite series be defined as a limit of partial sums?
6. What does it mean for an infinite series to converge?
7. What does it mean for an infinite series to diverge?
8. What is the relationship between the convergence of a series and the convergence of its sequence of partial sums?
9. What does it mean for a series to diverge to infinity?
10. How can you determine whether a series converges by examining its partial sums?
11. What is a geometric series? Write its general form.
12. What is the common ratio of a geometric series?
13. Under what condition does a geometric series converge?
14. What is the formula for the sum of a convergent geometric series?
15. What happens if the common ratio r of a geometric series satisfies $|r| \geq 1$?
16. How can you derive the formula for the sum of a geometric series?
17. What is a telescoping series?
18. How do terms cancel in a telescoping series?
19. How can you evaluate the sum of a telescoping series?
20. Why are telescoping series often easier to evaluate than other series?
21. Give an example of a telescoping series and describe its behavior.
22. What is the harmonic series? Write its general form.
23. Does the harmonic series converge or diverge?
24. Why is it not sufficient for $a_n \rightarrow 0$ to guarantee that $\sum a_n$ converges?
25. What is the divergence test (nth-term test for divergence)?
26. If $\lim_{n \rightarrow \infty} a_n \neq 0$, what can you conclude about $\sum a_n$?
27. If $\lim_{n \rightarrow \infty} a_n = 0$, what can you conclude about $\sum a_n$?
28. How are infinite series used in applications such as approximations or modeling?
29. How does the concept of an infinite series extend the idea of a finite sum?
30. How can infinite series be used to represent functions?
31. What is the intuitive meaning of “adding infinitely many terms”?
32. How does the order of terms affect the value of a finite sum? Does this idea carry over to infinite series?

3.3 Calculus Vol2 - 5.3 The Divergence and Integral Tests

<https://openstax.org/books/calculus-volume-2/pages/5-3-the-divergence-and-integral-tests>

1. Why is it often difficult to determine convergence of a series by directly evaluating partial sums?
2. What is the divergence test (nth-term test)?
3. State the divergence test formally.
4. Why does $\sum_{n=1}^{\infty} a_n$ converging imply that $a_n \rightarrow 0$?
5. If $\lim_{n \rightarrow \infty} a_n \neq 0$, what can you conclude about the series $\sum a_n$?
6. If $\lim_{n \rightarrow \infty} a_n = 0$, what can you conclude about the series $\sum a_n$?
7. Why is the divergence test only useful for proving divergence and not convergence?
8. Give an example of a series where $a_n \rightarrow 0$ but the series diverges.
9. How do you apply the divergence test in practice to a given series?
10. What kinds of limits of a_n guarantee divergence (nonzero limit, limit does not exist)?
11. What is the idea behind comparing a series to an improper integral?
12. What is the integral test?
13. What conditions must a function $f(x)$ satisfy to apply the integral test (positivity, continuity, monotonicity)?
14. How are the series $\sum_{n=1}^{\infty} a_n$ and the integral $\int_1^{\infty} f(x), dx$ related in the integral test?
15. State the integral test formally.
16. If the improper integral $\int_1^{\infty} f(x), dx$ converges, what can you conclude about the series?
17. If the improper integral diverges, what can you conclude about the series?
18. How can the integral test be used to prove that the harmonic series diverges?
19. What is a p -series? Write its general form.
20. For what values of p does the p -series $\sum_{n=1}^{\infty} \frac{1}{n^p}$ converge?
21. For what values of p does the p -series diverge?
22. How can the integral test be used to establish the convergence properties of p -series?
23. What is the remainder R_N of a series after N terms?
24. How can the integral test be used to estimate the remainder of a series?
25. State the inequality involving the remainder R_N and integrals.
26. How do upper and lower bounds for R_N arise from the integral test?
27. Why is it useful to estimate the error when approximating a series by partial sums?
28. How can you choose N to ensure the error is less than a desired tolerance?
29. What is the geometric interpretation of the integral test using areas under curves?
30. How does monotonicity of $f(x)$ play a role in bounding the series with integrals?
31. In what situations is the integral test particularly useful compared to other tests?

3.4 Calculus Vol2 - 5.4 Comparison Tests

<https://openstax.org/books/calculus-volume-2/pages/5-4-comparison-tests>

1. What is the main idea behind comparison tests for infinite series?
2. For what type of series (in terms of sign of terms) are comparison tests typically used?
3. Why are geometric series and p -series commonly used in comparison tests?
4. State the Direct Comparison Test for series $\sum a_n$ and $\sum b_n$ with positive terms.
5. If $0 \leq a_n \leq b_n$ and $\sum b_n$ converges, what can be concluded about $\sum a_n$?
6. If $0 \leq b_n \leq a_n$ and $\sum b_n$ diverges, what can be concluded about $\sum a_n$?
7. In which cases is the Direct Comparison Test inconclusive?
8. How do you choose an appropriate comparison series b_n for a given series a_n ?
9. Why is it useful to compare a series to a p -series of the form $\sum \frac{1}{n^p}$?
10. What is the behavior of the p -series $\sum \frac{1}{n^p}$ for different values of p ?
11. What is the behavior of a geometric series $\sum ar^n$ depending on the value of r ?
12. State the Limit Comparison Test for two series $\sum a_n$ and $\sum b_n$ with positive terms.
13. What does it mean if

$$\lim_{n \rightarrow \infty} \frac{a_n}{b_n} = c,$$

where $0 < c < \infty$?

14. What can be concluded if the limit in the Limit Comparison Test is 0?
15. What can be concluded if the limit in the Limit Comparison Test is ∞ ?
16. When is the Limit Comparison Test more useful than the Direct Comparison Test?
17. Why does the Limit Comparison Test work even when inequalities between a_n and b_n are hard to establish?
18. What are common functions or expressions you simplify when applying the Limit Comparison Test?
19. How do you simplify rational expressions in n when applying comparison tests?
20. How do logarithmic terms (e.g., $\ln n$) affect comparison with p -series?
21. How do roots (e.g., $\sqrt{n}$) affect comparison with p -series?
22. Give an example of a series that can be compared to $\sum \frac{1}{n^2}$ and explain why.
23. Give an example of a series that can be compared to $\sum \frac{1}{n}$ and explain why.
24. What is the general strategy for proving convergence using comparison tests?
25. What is the general strategy for proving divergence using comparison tests?
26. Why must terms a_n and b_n be positive for these tests?
27. Can comparison tests determine the exact sum of a series? Why or why not?
28. How do comparison tests relate to the Integral Test conceptually?
29. What role do asymptotic behaviors of functions play in the Limit Comparison Test?

30. How can you justify that two sequences have the same “growth rate” for comparison?
31. When comparing $\frac{1}{n^2+1}$ to $\frac{1}{n^2}$, why does the comparison work?

3.5 Calculus Vol2 - 5.5 Alternating Series

<https://openstax.org/books/calculus-volume-2/pages/5-5-alternating-series>

1. What is an alternating series? Write its general form.
2. How can an alternating series be written using the factor $(-1)^n$ or $(-1)^{n+1}$?
3. In the representation $\sum_{n=1}^{\infty} (-1)^{n+1} b_n$, what conditions are imposed on b_n ?
4. State the Alternating Series Test (Leibniz Test).
5. What two conditions must the sequence $\{b_n\}$ satisfy for the Alternating Series Test to apply?
6. Why is the condition $\lim_{n \rightarrow \infty} b_n = 0$ necessary for convergence?
7. Does the Alternating Series Test guarantee absolute convergence or only convergence? Explain.
8. Give an example of a series that converges by the Alternating Series Test.
9. What is meant by absolute convergence?
10. What is meant by conditional convergence?
11. What is the relationship between absolute convergence and convergence of a series?
12. Give an example of a series that converges conditionally but not absolutely.
13. How can you test whether an alternating series converges absolutely?
14. If $\sum |a_n|$ diverges but $\sum a_n$ converges, what type of convergence does the series have?
15. What is the alternating harmonic series? Does it converge absolutely or conditionally?
16. Define the n th partial sum S_N of a series.
17. What is the remainder R_N of a series?
18. For an alternating series satisfying the Alternating Series Test, what inequality bounds the remainder R_N ?
19. Write the inequality relating $|R_N|$ and b_{N+1} .
20. What does the remainder estimate tell us about the error when approximating a sum by S_N ?
21. How can you use the remainder estimate to determine how many terms are needed for a desired accuracy?
22. If you approximate the sum of an alternating series by S_N , what is the maximum possible error?
23. Why do partial sums of an alternating series tend to oscillate around the true sum?
24. Describe the behavior of even and odd partial sums in an alternating series that satisfies the test.
25. How can you estimate the sum of an alternating series to within a given tolerance ε ?
26. For the series $\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n^2}$, how would you bound the error after N terms?
27. What role does monotonicity ($b_{n+1} \leq b_n$) play in the Alternating Series Test?

28. Can an alternating series converge if the sequence b_n is not decreasing? What does the test say about this?
29. Compare the convergence behavior of $\sum \frac{(-1)^{n+1}}{n}$ and $\sum \frac{1}{n}$.
30. Why is it often difficult to compute the exact sum of an alternating series?
31. What practical advantage does the Alternating Series Estimation Theorem provide?
32. How does the Alternating Series Test differ from comparison or ratio tests?
33. When analyzing a series, why is it useful to check absolute convergence first?
34. What happens if $\lim_{n \rightarrow \infty} a_n \neq 0$ for an alternating series?

3.6 Calculus Vol2 - 5.6 Ratio and Root Tests

<https://openstax.org/books/calculus-volume-2/pages/5-6-ratio-and-root-tests>

1. Why is the condition $\lim_{n \rightarrow \infty} a_n = 0$ not sufficient to guarantee convergence of $\sum a_n$?
2. Give an example of two series where $a_n \rightarrow 0$ in both cases, but one converges and the other diverges.
3. What does it mean for the terms of a series to approach zero “fast enough”?
4. State the Ratio Test for a series $\sum a_n$.
5. Define

$$\rho = \lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right|.$$

6. What conclusion can be drawn if $\rho < 1$ in the Ratio Test?
7. What conclusion can be drawn if $\rho > 1$ (or $\rho = \infty$) in the Ratio Test?
8. What happens if $\rho = 1$ in the Ratio Test?
9. What type of convergence does the Ratio Test establish when $\rho < 1$?
10. For what kinds of series is the Ratio Test especially useful?
11. Apply the Ratio Test conceptually: what happens to the ratio $\left| \frac{a_{n+1}}{a_n} \right|$ for factorial-type expressions?
12. State the Root Test for a series $\sum a_n$.
13. Define

$$\rho = \lim_{n \rightarrow \infty} \sqrt[n]{|a_n|}.$$

14. What conclusion can be drawn if $\rho < 1$ in the Root Test?
15. What conclusion can be drawn if $\rho > 1$ in the Root Test?
16. What happens if $\rho = 1$ in the Root Test?
17. Compare the Ratio Test and Root Test: what is similar about their conclusions?
18. In what situations is the Root Test particularly useful?
19. How does the Root Test behave for expressions involving powers like $(b_n)^n$?
20. What is absolute convergence, and how do these tests relate to it?
21. Why do both the Ratio and Root Tests use absolute values?
22. How are these tests useful for power series?
23. When should you consider using the Ratio Test over other tests?
24. When should you consider using the Root Test over other tests?
25. Why are these tests often easier to apply than comparison tests?
26. Can the Ratio or Root Test determine conditional convergence?
27. How do these tests relate to geometric series?

4 Calculus Vol2 - 6. Power Series

<https://openstax.org/books/calculus-volume-2/pages/6-introduction>

4.1 Calculus Vol2 - 6.1 Power Series and Functions

<https://openstax.org/books/calculus-volume-2/pages/6-1-power-series-and-functions>

1. What is the general form of a power series centered at $x = 0$?
2. Write the general form of a power series centered at $x = a$.
3. Explain in your own words why a power series can be thought of as an “infinite polynomial.”
4. Give two examples of power series centered at $x = 0$ with explicit formulas for their coefficients c_n .
5. What does it mean for a power series to be “centered at” a specific point?
6. State the three possible convergence behaviors of a power series centered at $x = a$.
7. Define the *interval of convergence* for a power series.
8. Define the *radius of convergence* of a power series.
9. Why does every power series always converge at its center?
10. Using the geometric series

$$\sum_{n=0}^{\infty} x^n,$$

determine its interval and radius of convergence.

11. How can the Ratio Test be used to help find the radius of convergence of a power series?
12. True/False: If a power series converges for some value x_0 , then it converges for all x with $|x - a| < |x_0 - a|$. Explain your answer.
13. What is the basic idea behind representing a given function $f(x)$ with a power series?
14. Give an example of a function that can be expressed with a power series based on geometric series manipulation.
15. How would you represent the function $\frac{1}{1-x}$ as a power series?
16. How would you represent the function $\frac{1}{1+x^2}$ as a power series?
17. Explain the relationship between power series and functions that are infinitely differentiable near their center.
18. For a power series with finite radius of convergence R , what can be said about convergence at the endpoints $x = a \pm R$?
19. How are the endpoints of the interval of convergence typically tested if the Ratio Test is inconclusive?
20. What implications does the convergence of a power series have for the differentiability and integrability of the represented function on its interval of convergence?

4.2 Calculus Vol2 - 6.2 Properties of Power Series

<https://openstax.org/books/calculus-volume-2/pages/6-2-properties-of-power-series>

1. What is the general result about combining two power series

$$\sum_{n=0}^{\infty} c_n x^n \quad \text{and} \quad \sum_{n=0}^{\infty} d_n x^n$$

that converge on the same interval I ? State the conditions for addition, scalar multiplication, and multiplication by a power of x .

2. Suppose two power series converge to functions $f(x)$ and $g(x)$ on an interval I . What function does the series

$$\sum_{n=0}^{\infty} (c_n + d_n) x^n$$

converge to on I ?

3. For a power series $\sum_{n=0}^{\infty} c_n x^n$ that converges to $f(x)$ on an interval I , what is the power series that converges to $b x^m f(x)$ for a real number b and integer $m \geq 0$?
4. If $\sum_{n=0}^{\infty} c_n x^n$ converges to $f(x)$ on an interval I , what is the new series that represents $f(b x^m)$ when $|b x^m|$ is in the interval I ?
5. What theorem guarantees that a power series can be differentiated and integrated term-by-term on its interval of convergence? What does this theorem say about the resulting series and their convergence?
6. Given a power series

$$f(x) = \sum_{n=0}^{\infty} c_n (x - a)^n,$$

write the term-by-term differentiated series for $f'(x)$ in summation form.

7. Given the same $f(x)$ above, write the term-by-term integrated series representing an antiderivative of $f(x)$.
8. Does term-by-term differentiation or integration change the radius of convergence of a power series? Explain and describe what can happen at the endpoints.
9. Using the known geometric series for $\frac{1}{1-x}$, what is the power series representation of $\frac{1}{(1-x)^2}$ found by term-by-term differentiation? Specify its interval of convergence and endpoint behavior.
10. Find a power series representation for $f(x) = \ln(1+x)$ by integrating the power series for f' and find its interval of convergence.
11. Find a power series representation for $f(x) = \tan^{-1}(x)$ by integrating the power series for f' and find its interval of convergence.

4.3 Calculus Vol2 - 6.3 Taylor and Maclaurin Series

<https://openstax.org/books/calculus-volume-2/pages/6-3-taylor-and-maclaurin-series>

1. What is the general form of the Taylor series for a function $f(x)$ centered at $x = a$? Write it using summation notation.
2. How does the Taylor series simplify when the center $a = 0$? What name is given to this special case?
3. For a function $f(x)$ with derivatives of all orders at $x = a$, write the explicit expression for its n -th Taylor polynomial $p_n(x)$ centered at a .
4. Explain why the coefficients of a power series representation of $f(x)$ at $x = a$ must be given by derivatives of f evaluated at a .
5. What condition must hold for a function's Taylor series to equal the original function $f(x)$ on an interval? Define the remainder $R_n(x)$ in this context.
6. State the Uniqueness of Taylor Series theorem: if a power series converges to $f(x)$ on an interval containing a , what can be said about that series?
7. Define the remainder term $R_n(x)$ for a Taylor polynomial approximation and write the formula for the Lagrange form of the remainder.
8. What does Taylor's Theorem with remainder tell us about the error when approximating $f(x)$ using its n -th Taylor polynomial?
9. Given a specific function $f(x)$, describe the steps to find the Maclaurin polynomial of degree n .
10. Find the Maclaurin series for e^x by writing out several derivatives at 0 and using the definition.
11. Find the Maclaurin series for $\sin x$ and $\cos x$. Explain the pattern in the derivatives that allows you to write the general term.
12. How can the ratio test be used to determine the radius of convergence for a Taylor or Maclaurin series?
13. For a given Taylor series, how would you check whether it actually converges to $f(x)$ (not just converges in value)?
14. Suppose you are given a function $f(x)$ with known derivatives at $x = a$. Write a question asking you to compute the 2nd and 3rd Taylor polynomials explicitly and compare them to the function near a .
15. Create a question that asks you to estimate the error in using the 3rd Taylor polynomial to approximate a function value at a given point using the remainder bound.
16. Why does the remainder $R_n(x) \rightarrow 0$ as $n \rightarrow \infty$ matter for Taylor series convergence?
17. Provide the Maclaurin series expansion for $\ln(1 + x)$ and specify its interval of convergence.
18. For a given nonzero center a , how would you find the Taylor series of $\ln(x)$ about $x = a$ and determine its interval of convergence?
19. Write a practice question that asks for the Taylor series of a non-elementary function (e.g., e^{-x^2}) and discuss whether it converges to the function.
20. Formulate a question about how to bound the remainder $R_n(x)$ using information about the maximum value of a derivative on an interval.

21. Write a question that asks you to use Taylor's theorem to prove that the Maclaurin series for e^x converges for all real x .
22. Create a problem asking you to approximate a definite integral using a Taylor or Maclaurin series where the antiderivative is non-elementary.

4.4 Calculus Vol2 - 6.4 Working with Taylor Series

<https://openstax.org/books/calculus-volume-2/pages/6-4-working-with-taylor-series>

1. What is the Taylor series of a function f centered at $x = a$?
2. What is the Maclaurin series? How does it relate to the Taylor series?
3. Write the Taylor series of f centered at a using sigma notation.
4. What is the n th Taylor polynomial $T_n(x)$ of a function f centered at a ?
5. What is the relationship between the Taylor polynomial $T_n(x)$ and the full Taylor series?
6. Compute the Maclaurin series of e^x .
7. Compute the Maclaurin series of $\sin x$.
8. Compute the Maclaurin series of $\cos x$.
9. Compute the Maclaurin series of $\frac{1}{1-x}$ and state its interval of convergence.
10. How can you obtain the Maclaurin series for $\ln(1+x)$ from the geometric series?
11. How can you obtain the Maclaurin series for $\arctan x$?
12. Given a known power series, how can you:
 - (a) Differentiate it term-by-term?
 - (b) Integrate it term-by-term?
13. What happens to the radius of convergence when you differentiate or integrate a power series?
14. How can you construct the Taylor series for e^{2x} using the known series for e^x ?
15. How can you construct the Taylor series for $\sin(3x)$?
16. How can you construct the Taylor series for xe^x ?
17. How can you construct the Taylor series for $\frac{x}{1-x^2}$?
18. How do you find the Taylor series of a function centered at a point $a \neq 0$?
19. Find the Taylor series of $\ln x$ centered at $x = 1$.
20. What substitution allows you to rewrite $\ln x$ in terms of $\ln(1+u)$?
21. How can you approximate a function using its Taylor polynomial?
22. What is Taylor's theorem?
23. What is the remainder term $R_n(x)$ in Taylor's theorem?
24. State the Lagrange form of the remainder.
25. How can you use the remainder term to bound the error of a Taylor approximation?
26. Suppose $|f^{(n+1)}(x)| \leq M$ on an interval containing a and x . What inequality bounds $|R_n(x)|$?
27. How do you determine how large n must be to guarantee a given approximation error?
28. Use a Taylor polynomial to approximate $e^{0.1}$. How can you estimate the error?
29. Use a Taylor polynomial to approximate $\sin(0.2)$. How can you estimate the error?

30. What does it mean for a function to be equal to its Taylor series?
31. Give an example of a function whose Taylor series converges but does not equal the function.
32. Why does the Taylor series of e^x converge for all x ?
33. How does the factorial in the denominator affect convergence of Taylor series?
34. When approximating near $x = a$, why is it better to center the Taylor polynomial at a rather than at 0?

5 Calculus Vol2 - 7. Parametric Equations and Polar Coordinates

<https://openstax.org/books/calculus-volume-2/pages/7-introduction>

5.1 Calculus Vol2 - 7.1 Parametric Equations

<https://openstax.org/books/calculus-volume-2/pages/7-1-parametric-equations>

1. What is a parametric equation, and how does it differ from a rectangular equation $y = f(x)$?
2. What is a parameter, and what does it typically represent in applications (e.g., time)?
3. How are x and y expressed in terms of a parameter t in parametric form?
4. How does a point $(x(t), y(t))$ move as the parameter t increases?
5. Why are parametric equations useful for describing curves that are not functions?
6. What does the orientation of a parametric curve mean, and how is it determined?
7. Why can the same curve have multiple different parametric representations?
8. How can you eliminate the parameter to convert parametric equations into a rectangular equation?
9. What is the purpose of eliminating the parameter?
10. How do restrictions on the parameter t affect the portion of the curve that is traced?
11. What are the parametric equations of a circle, and how do they relate to trigonometric functions?
12. How can you verify that a parametric curve represents a known geometric shape?
13. How do parametric equations provide more information than a rectangular equation (e.g., direction of motion)?
14. What does it mean for a curve to be traced once, multiple times, or partially?
15. How is the derivative $\frac{dy}{dx}$ computed for a parametric curve?
16. Why is the formula $\frac{dy}{dx} = \frac{dy/dt}{dx/dt}$ valid?
17. What happens when $\frac{dx}{dt} = 0$ in the formula for $\frac{dy}{dx}$?
18. How can you find points where the tangent line is horizontal or vertical?
19. What is a critical point for a curve defined parametrically?
20. How can you find the equation of a tangent line to a parametric curve at a given value of t ?
21. How do you compute $\frac{dx}{dt}$ and $\frac{dy}{dt}$ for given parametric equations?
22. What is the relationship between parametric equations and vector-valued functions?
23. How can parametric equations describe curves that cannot be written as $y = f(x)$?

5.2 Calculus Vol2 - 7.2 Calculus of Parametric Curves

<https://openstax.org/books/calculus-volume-2/pages/7-2-calculus-of-parametric-curves>

1. How do you compute the derivative $\frac{dy}{dx}$ for a parametric curve defined by $x(t)$ and $y(t)$?
2. Why is the formula $\frac{dy}{dx} = \frac{dy/dt}{dx/dt}$ valid, and what assumption is required about $\frac{dx}{dt}$?
3. How do you find the equation of the tangent line to a parametric curve at a specific value $t = t_0$?
4. How can you determine where a parametric curve has a horizontal tangent?
5. How can you determine where a parametric curve has a vertical tangent?
6. What happens to $\frac{dy}{dx}$ when $\frac{dx}{dt} = 0$?
7. How do you compute the second derivative $\frac{d^2y}{dx^2}$ for a parametric curve?
8. Why is the second derivative useful when analyzing concavity of parametric curves?
9. How can you interpret parametric equations as describing motion of a particle?
10. What are the velocity components of a particle moving along a parametric curve?
11. How is speed related to $x'(t)$ and $y'(t)$?
12. How do you compute the arc length of a parametric curve between $t = a$ and $t = b$?
13. What is the formula $L = \int_a^b \sqrt{(x'(t))^2 + (y'(t))^2} dt$ and how is it derived conceptually?
14. How do you compute the area under a parametric curve?
15. Why is the formula $A = \int_{t_1}^{t_2} y(t) x'(t) dt$ used for area?
16. How does the sign of $x'(t)$ affect the computed area?
17. What conditions must hold for the area formula for parametric curves to be valid?
18. How do you handle curves that are traced more than once when computing area?
19. How can you determine whether a parametric curve is traced once or multiple times?
20. How do you compute the surface area generated by revolving a parametric curve about the x -axis?
21. What is the formula $S = 2\pi \int_a^b y(t) \sqrt{(x'(t))^2 + (y'(t))^2} dt$ and when is it used?
22. How does the surface area formula change when revolving around the y -axis?
23. What geometric idea underlies the surface area of revolution for parametric curves?

5.3 Calculus Vol2 - 7.3 Polar Coordinates

<https://openstax.org/books/calculus-volume-2/pages/7-3-polar-coordinates>

1. What is the polar coordinate system, and how does it differ from the Cartesian coordinate system?
2. What do the coordinates (r, θ) represent in the polar coordinate system?
3. What is the geometric meaning of the radial coordinate r ?
4. What is the geometric meaning of the angle θ ?
5. What are the pole and the polar axis?
6. How are positive and negative angles interpreted in polar coordinates?
7. What happens when the radial coordinate r is negative?
8. Why does a single point in the plane have infinitely many polar representations?
9. How can the same point be represented using (r, θ) and $(r, \theta + 2\pi)$?
10. How can a point be represented using a negative radius and a shifted angle?
11. What are the conversion formulas from polar to rectangular coordinates?
12. What are the conversion formulas from rectangular to polar coordinates?
13. How can you graph a polar equation $r = f(\theta)$?
14. Why are polar coordinates especially useful for describing circular or radial patterns?
15. What types of curves are naturally expressed in polar coordinates (e.g., circles, spirals)?
16. What does it mean for a polar graph to have symmetry about the pole?
17. What does it mean for a polar graph to have symmetry about the line $\theta = \frac{\pi}{2}$?
18. How can symmetry help simplify graphing polar equations?
19. What is a polar equation, and how does it define a curve?
20. How do you interpret the equation $r = c$ geometrically?
21. How do you interpret the equation $\theta = c$ geometrically?

5.4 Calculus Vol2 - 7.4 Area and Arc Length in Polar Coordinates

<https://openstax.org/books/calculus-volume-2/pages/7-4-area-and-arc-length-in-polar-coordinates>

1. How is the concept of area in polar coordinates different from area in rectangular coordinates?
2. Why are sectors of circles used instead of rectangles in deriving area formulas in polar coordinates?
3. How do you approximate the area of a polar region using a Riemann sum?
4. What is the formula for the area of a region defined by $r = f(\theta)$ on $[\alpha, \beta]$?
5. Why does the factor $\frac{1}{2}$ appear in the polar area formula?
6. What geometric quantity does $(f(\theta))^2$ represent in the area formula?
7. How do you compute the area enclosed by a single polar curve?
8. How do you find the area between two polar curves?
9. Why must you find points of intersection when computing area between two polar curves?
10. How do you determine which curve is the outer curve and which is the inner curve?
11. What happens if a polar curve is traced more than once when computing area?
12. How can symmetry be used to simplify area calculations in polar coordinates?
13. What is the geometric interpretation of the integral $\frac{1}{2} \int_{\alpha}^{\beta} r^2, d\theta$?
14. How is arc length defined for a polar curve $r = f(\theta)$?
15. Why does the arc length formula involve both r and $\frac{dr}{d\theta}$?
16. How can the arc length formula be derived conceptually from parametric equations?
17. What is the relationship between polar coordinates and parametric form when computing arc length?
18. How do you compute $\frac{dr}{d\theta}$ for a given polar function?
19. Why does the expression $\sqrt{r^2 + \left(\frac{dr}{d\theta}\right)^2}$ represent speed along the curve?
20. How can you verify that a polar curve encloses a finite area?
21. What are the key similarities and differences between area and arc length formulas in rectangular, parametric, and polar forms?
22. Why are polar coordinates particularly useful for computing areas of radially symmetric regions?
23. What types of curves (e.g., cardioids, roses, spirals) are naturally analyzed using these formulas?

5.5 Calculus Vol2 - 7.5 Conic Sections

<https://openstax.org/books/calculus-volume-2/pages/7-5-conic-sections>

1. What is a conic section, and how is it formed geometrically?
2. What are the four main types of conic sections?
3. How does slicing a cone at different angles produce different conic sections?
4. What is the definition of a conic section in terms of a focus and a directrix?
5. What is a focus, and how is it used to define a conic section?
6. What is a directrix, and what role does it play in defining conics?
7. What is eccentricity, and how is it defined as a ratio of distances?
8. How does the value of eccentricity determine the type of conic (circle, ellipse, parabola, hyperbola)?
9. What is the eccentricity of a circle, ellipse, parabola, and hyperbola?
10. What is the geometric definition of a parabola in terms of focus and directrix?
11. What is the standard equation of a parabola, and what do its parameters represent?
12. How does the parameter p relate to the vertex and focus of a parabola?
13. What is the geometric definition of an ellipse?
14. What is the standard equation of an ellipse centered at (h, k) ?
15. What are the major and minor axes of an ellipse?
16. How are the quantities a , b , and c related in an ellipse?
17. Where are the foci located in an ellipse?
18. What is the geometric definition of a hyperbola?
19. What is the standard equation of a hyperbola centered at (h, k) ?
20. How are the quantities a , b , and c related in a hyperbola?
21. Where are the foci and vertices located in a hyperbola?
22. What are asymptotes, and why are they important for hyperbolas?
23. What is the general second-degree equation of a conic section?
24. What is the discriminant $4AC - B^2$, and how is it used to classify conics?
25. How can you identify a conic section from its general quadratic equation?
26. How can you convert a general conic equation into standard form?
27. What role does completing the square play in rewriting conic equations?
28. How are conic sections represented in polar coordinates?
29. What is the polar form of a conic section involving eccentricity e and parameter p ?
30. How does the equation $r = \frac{ep}{1 \pm e \cos \theta}$ or $r = \frac{ep}{1 \pm e \sin \theta}$ describe conics?
31. What is the focal parameter p , and what is its geometric meaning?
32. How can you determine the type of conic from its polar equation?

33. What are the key differences between conics in rectangular form and polar form?
34. How does changing parameters in the equations affect the shape and position of a conic?